

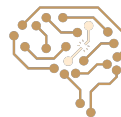
Decision Tree

Arash Marioriyad

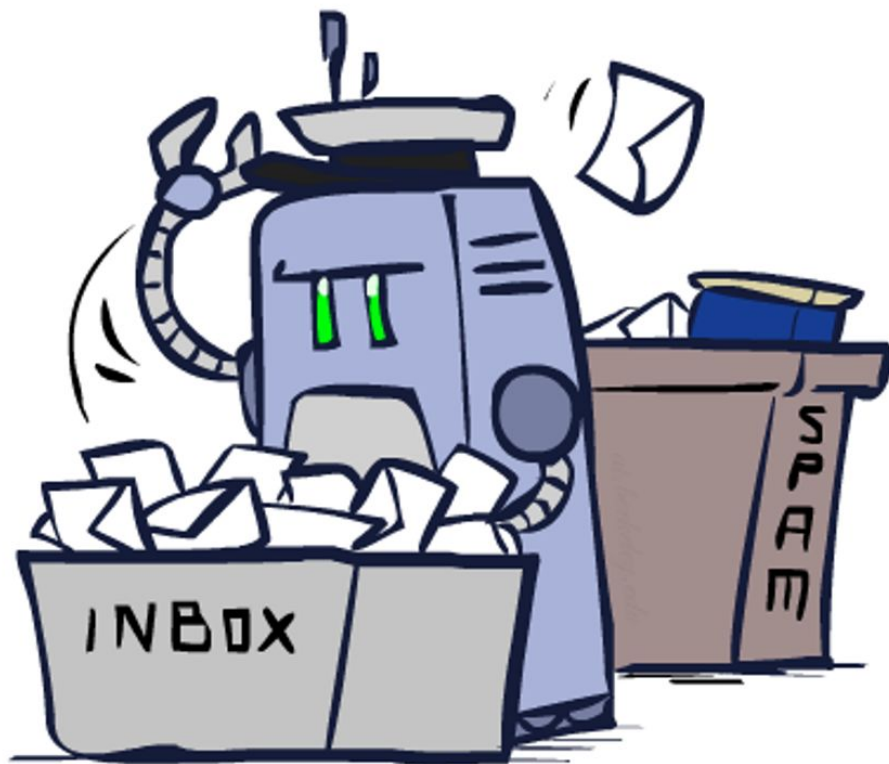
3 Khordad 1405

CE 417-Artificial Intelligence
Sharif Univ. of Tech.

Most Slides have been adopted from MIT CSAIL Lab
slides & Sharif UT CE417



Classification



Example: Spam Filter

- **Input:** an email
- **Output:** “spam” or “ham”
- **Data:** large collection of example emails, each labeled “spam” or “ham”
- **Note:** someone has to hand label all this data!
- **Goal:** to learn a model using the data, to predict labels of new, future emails
- **Features:** The attributes (words, text patterns, sender information, etc.,) that contribute in making an email ham or spam



Dear Sir.

First, I must solicit your confidence in this transaction, this is by virtue of its nature as being utterly confidential and top secret. ...



TO BE REMOVED FROM FUTURE MAILINGS, SIMPLY REPLY TO THIS MESSAGE AND PUT "REMOVE" IN THE SUBJECT.

99 MILLION EMAIL ADDRESSES FOR ONLY \$99



Ok, I know this is blatantly OT but I'm beginning to go insane. Had an old Dell Dimension XPS sitting in the corner and decided to put it to use, I know it was working pre being stuck in the corner, but when I plugged it in, hit the power nothing happened.



Example: Digit Recognition

- **Input:** images (pixels)
- **Output:** a digit 0-9
- **Data:** a large collection of example images, each labeled with a digit
- **Note:** someone has to hand label all this data!
- **Goal:** to learn a model using data, to predict labels of new, future digit images
- **Features:** The attributes used to make the digit decision
 - Pixels: example (6,8)=ON
 - Shape Patterns



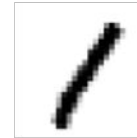
0



1



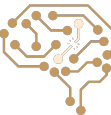
2



1

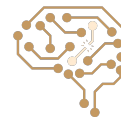
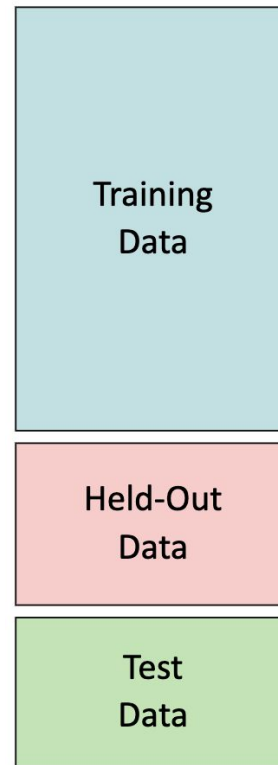


??



Data

- Three Types of Datasets (Labeled Instances):
 - Training Set
 - Validation (Held-out) Set
 - Test Set



Bayes Net For Classification: Nodes

- **Class node (Y):**
 - Represents the target label or category we want to predict.
- **Feature (attribute) nodes ($X_1, X_2, \dots, X_n$):**
 - Observed variables describing the instance
- Sometimes feature nodes are shown as $F_1, F_2, \dots, F_n$



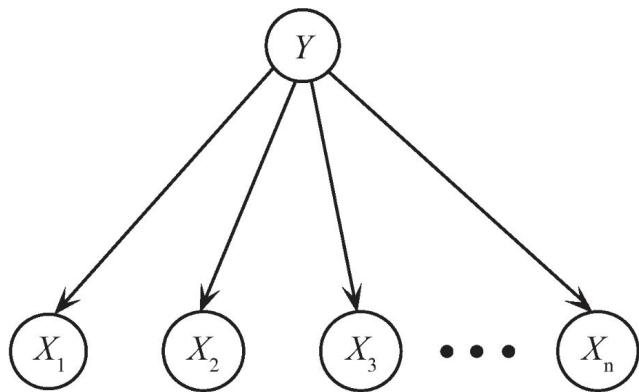
Bayes Net For Classification: Edges

- **Edges between class node and features:**
 - $Y \rightarrow X_i$: class causes the observed attributes.
- **Dependency edges among features:**
 - Edges between X_i and X_j to encode relations between features

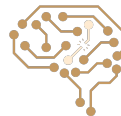


Naive Bayes

- Naive Assumption: all features are **independent** of each other
- Edges are only from the class variable to the feature variables



$$P(Y, X_1, \dots, X_n) = P(Y) \prod_{i=1}^n P(X_i | Y)$$



Naive Bayes Inference

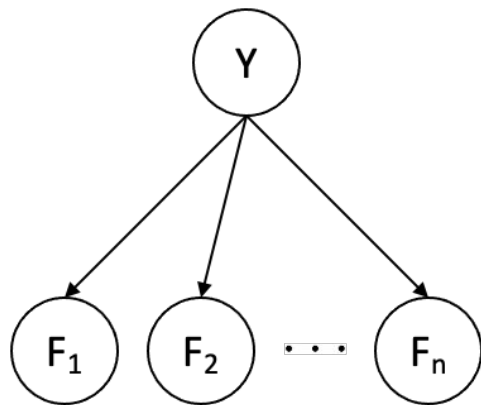
- A Naive Bayes model:

$|Y|$ parameters

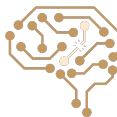
$$P(Y, F_1 \dots F_n) = P(Y) \prod_i P(F_i | Y)$$

$|Y| \times |F|^n$ values

$n \times |F| \times |Y|$
parameters



- We only have to specify how each feature depends on the class
- Total number of parameters is linear in n
- Model is very simplistic, but often works anyway



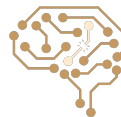
Bayes Net For Classification: Goal

- Use a probabilistic graphical model to compute:

$$\arg \max_y P(Y = y \mid x_1, \dots, x_n) = \arg \max_y \frac{P(Y = y, x_1, \dots, x_n)}{\sum_{y'} P(Y = y', x_1, \dots, x_n)}.$$

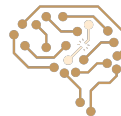
- And the joint distribution is:

$$P(Y, X_1, \dots, X_n) = P(Y) \prod_{i=1}^n P(X_i \mid \text{Pa}(X_i))$$



What Must We Learn in Naïve Bayes?

- Estimates of local conditional probability tables:
 - $P(Y)$, the prior over labels
 - $P(F_i|Y)$ for each feature (evidence variable)
- These probabilities are collectively called the parameters of the model and denoted by θ
- Up until now, we assumed these appeared by magic, but...
- But they typically come from training data



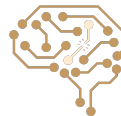
How to Learn Parameters From Data?

- Algorithm: **Maximum Likelihood Estimation (MLE)**
- Two types of parameters in Naive Bayes
- **Class Prior:**

$$P(Y = y) = \frac{\text{number of training examples with } Y = y}{N}$$

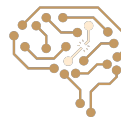
- **Conditional Probabilities:**

$$P(X_i = x \mid Y = y) = \frac{\#\{X_i = x, Y = y\}}{\#\{Y = y\}}$$



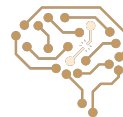
Overfitting

- Relative frequency parameters will overfit the training data!
 - Just because we never saw a 3 with pixel (15,15) on during training doesn't mean we won't see it at test time
 - Unlikely that every occurrence of "minute" is 100% spam
 - Unlikely that every occurrence of "seriously" is 100% ham
 - What about all the words that don't occur in the training set at all?
 - In general, we can't go around giving unseen events zero probability



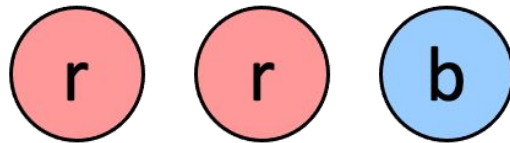
Generalization

- Perform well on unseen data, not just the training examples
- Failure mode = Overfitting
 - the model fits training frequencies too closely and assigns unreliable probabilities to new cases
- Solution in Naive Bayes: Smoothing



Laplace Smoothing in General

- Pretend you saw every outcome k extra times



$$P_{LAP,k}(x) = \frac{c(x) + k}{N + k|X|}$$

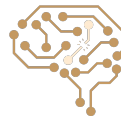
- What's Laplace with $k = 0$?
- k is the strength of the prior
- Laplace for conditionals:

$$P_{LAP,0}(X) = \left\langle \frac{2}{3}, \frac{1}{3} \right\rangle$$

$$P_{LAP,1}(X) = \left\langle \frac{3}{5}, \frac{2}{5} \right\rangle$$

$$P_{LAP,100}(X) = \left\langle \frac{102}{203}, \frac{101}{203} \right\rangle$$

$$P_{LAP,k}(x|y) = \frac{c(x,y) + k}{c(y) + k|X|}$$



Today: Decision Tree

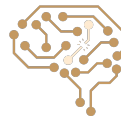


Example: Predict Fuel Efficiency (miles per gallon)

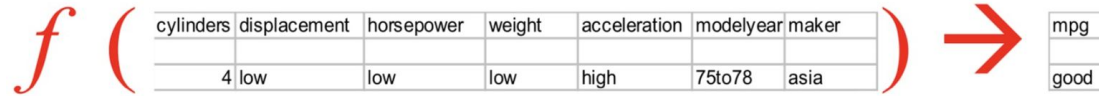
- 40 data points
- Goal: predict MPG
- Need to find:
 $f : X \rightarrow Y$
- Discrete data (for now)

mpg	cylinders	displacement	horsepower	weight	acceleration	modelyear	maker
good	4	low	low	low	high	75to78	asia
bad	6	medium	medium	medium	medium	70to74	america
bad	4	medium	medium	medium	low	75to78	europa
bad	8	high	high	high	low	70to74	america
bad	6	medium	medium	medium	medium	70to74	america
bad	4	low	medium	low	medium	70to74	asia
bad	4	low	medium	low	low	70to74	asia
bad	8	high	high	high	low	75to78	america
:	:	:	:	:	:	:	:
:	:	:	:	:	:	:	:
:	:	:	:	:	:	:	:
bad	8	high	high	high	low	70to74	america
good	8	high	medium	high	high	79to83	america
bad	8	high	high	high	low	75to78	america
good	4	low	low	low	low	79to83	america
bad	6	medium	medium	medium	high	75to78	america
good	4	medium	low	low	low	79to83	america
good	4	low	low	medium	high	79to83	america
bad	8	high	high	high	low	70to74	america
good	4	low	medium	low	medium	75to78	europa
bad	5	medium	medium	medium	medium	75to78	europa

Y
 X



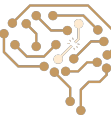
Finding Hypothesis



General Propositional Logic?

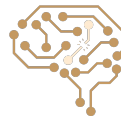
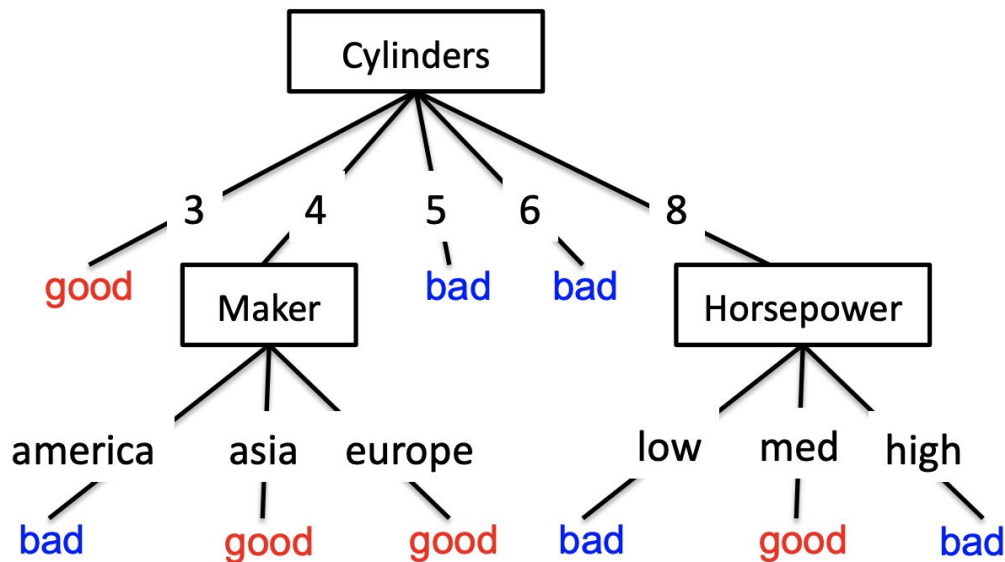
maker=asia $\vee$ weight=low

Need to find "Hypothesis": $f : X \rightarrow Y$



Hypotheses: decision trees $f : X \rightarrow Y$

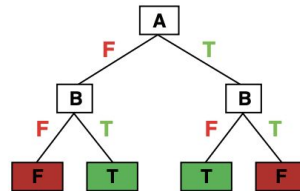
- Each internal node tests an attribute x_i
- Each branch assigns an attribute value $x_i = v$
- Each leaf assigns a class y
- To classify input x : traverse the tree from root to leaf, output the labeled y



What functions can be represented?

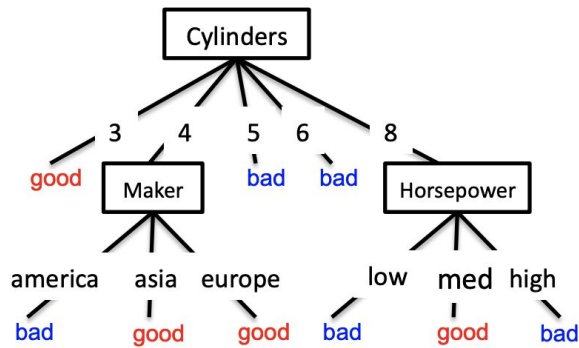
- Decision trees can represent any function of the input attributes!

A	B	A xor B
F	F	F
F	T	T
T	F	T
T	T	F

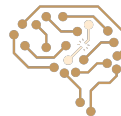


(Figure from Stuart Russell)

- For Boolean functions, path to leaf gives truth table row
- But, could require exponentially many nodes...



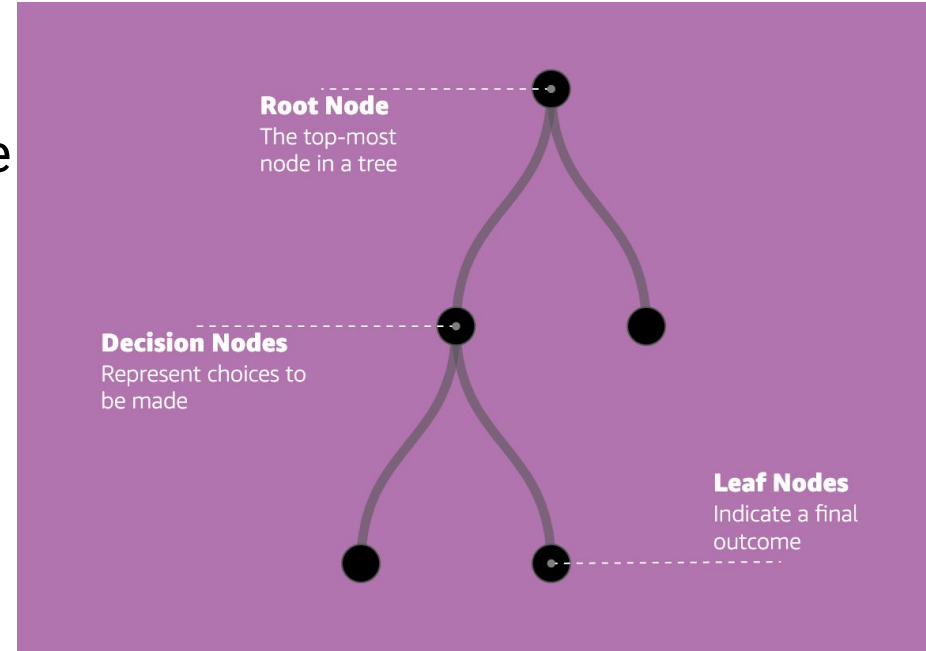
$\text{cyl}=3 \vee (\text{cyl}=4 \wedge (\text{maker}=\text{asia} \vee \text{maker}=\text{europe})) \vee \dots$



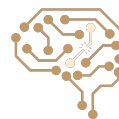
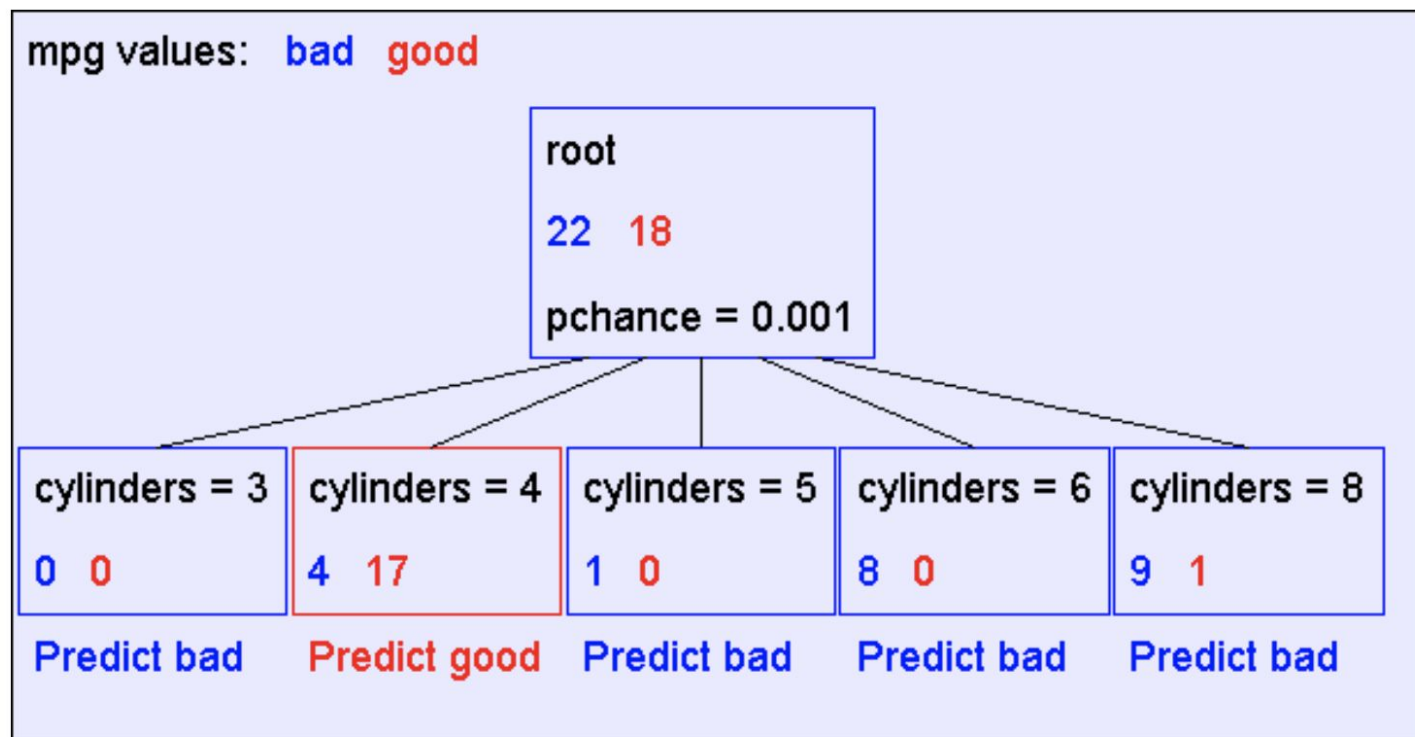
How to choose the best tree?

Let's first look at how to split nodes,
then consider how to find the best tree

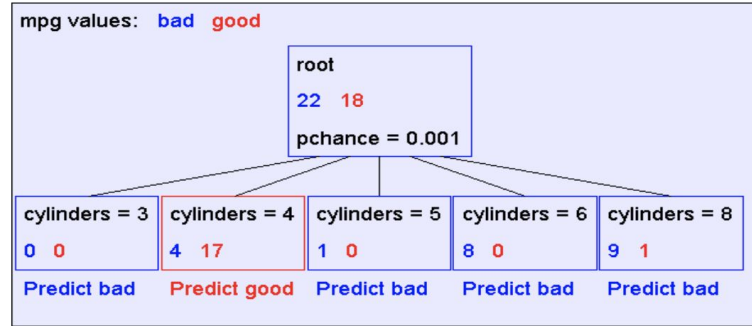
Finding the best tree is a search
problem!



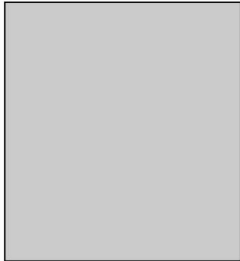
Building a simple tree



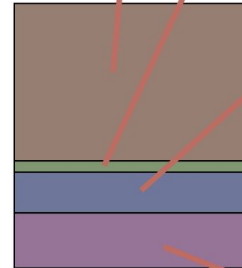
Recursive Step



Take the
Original
Dataset..



And partition it
according
to the value of
the attribute we
split on



Records
in which
cylinders
= 4

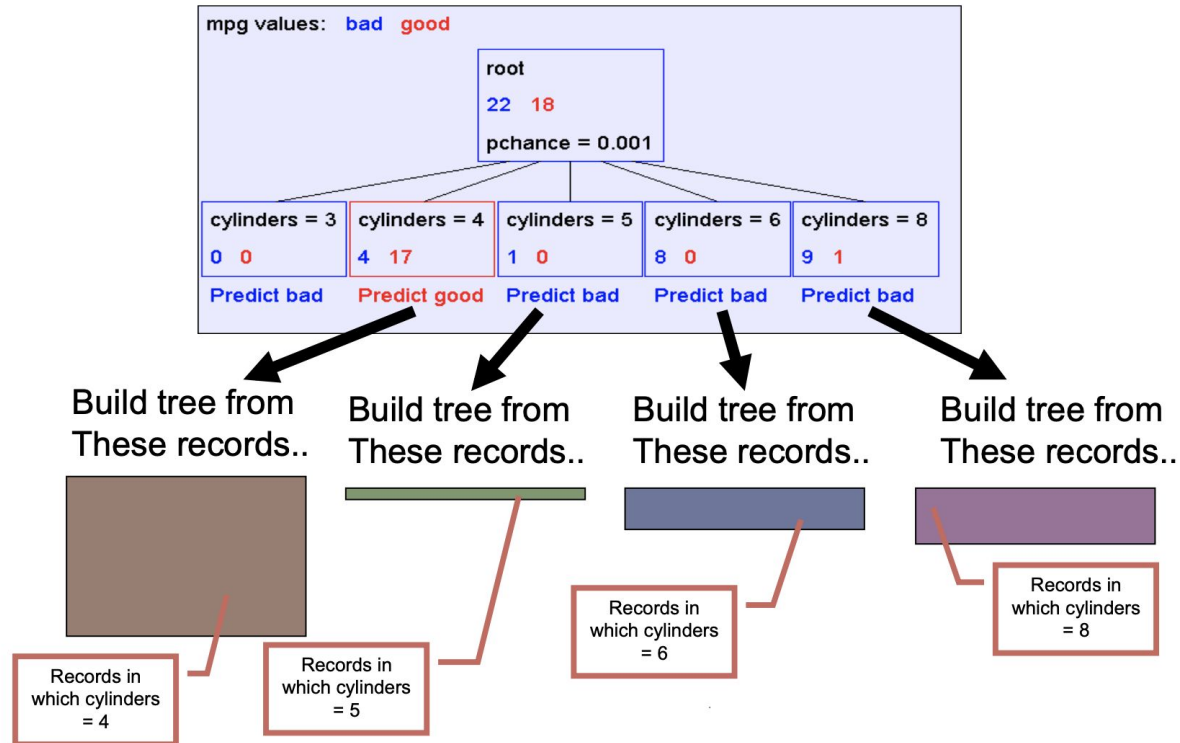
Records
in which
cylinders
= 5

Records
in which
cylinders
= 6

Records
in which
cylinders
= 8

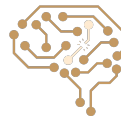


Recursive Step (cont.)



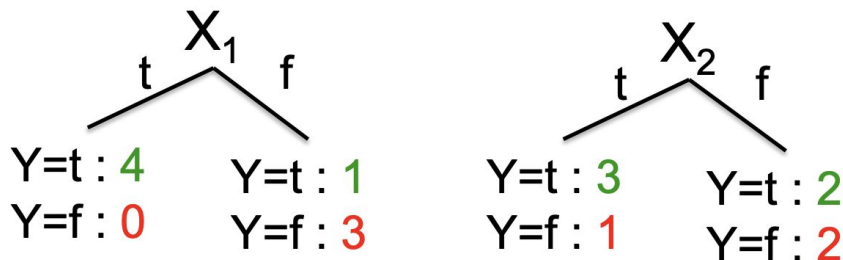
Learning Decision Trees is Hard!

- Learning the simplest (smallest) decision tree is a NP-complete problem
- Resort to a greedy heuristic:
 - Start from empty decision tree
 - Split on next best attribute (feature)
 - Recurse
- Which attribute gives the best split?
- When to stop recursion?



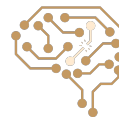
Choosing a Good Attribute

Would we prefer to split on X_1 or X_2 ?



X_1	X_2	Y
T	T	T
T	F	T
T	T	T
T	F	T
F	T	T
F	F	F
F	T	F
F	F	F

Idea: use counts at leaves to define probability distributions, so we can measure uncertainty!

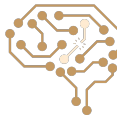


Measuring uncertainty

- Good split if we are more certain about classification after split
 - Deterministic good (all true or all false)
 - Uniform distribution bad
 - What about distributions in between?

$P(Y=A) = 1/2$	$P(Y=B) = 1/4$	$P(Y=C) = 1/8$	$P(Y=D) = 1/8$
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$P(Y=A) = 1/4$	$P(Y=B) = 1/4$	$P(Y=C) = 1/4$	$P(Y=D) = 1/4$
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Entropy

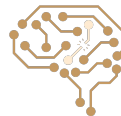
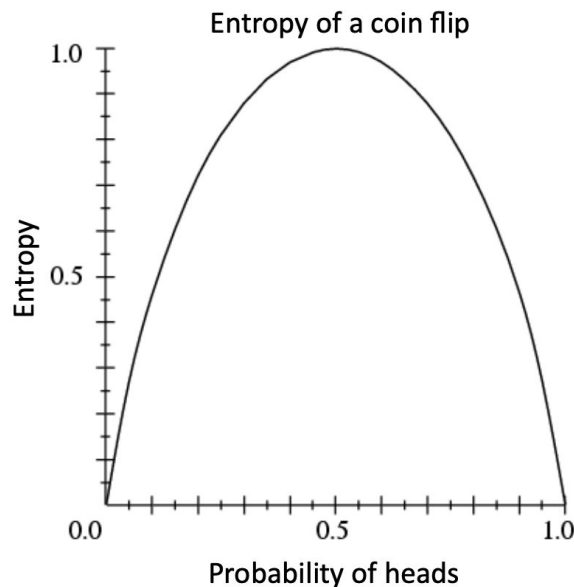
Entropy $H(Y)$ of a random variable Y

$$H(Y) = - \sum_{i=1}^k P(Y = y_i) \log_2 P(Y = y_i)$$

More uncertainty, more entropy!

Information Theory interpretation:

$H(Y)$ is the expected number of bits needed to encode a randomly drawn value of Y (under most efficient code)



High vs Low Entropy

- “High Entropy”
 - Y is from a uniform like distribution
 - Flat histogram
 - Values sampled from it are less predictable
- “Low Entropy”
 - Y is from a varied (peaks and valleys) distribution
 - Histogram has many lows and highs
 - Values sampled from it are more predictable



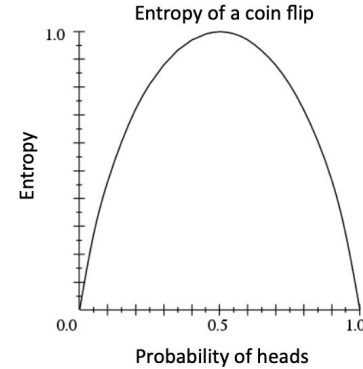
Entropy Example

$$H(Y) = - \sum_{i=1}^k P(Y = y_i) \log_2 P(Y = y_i)$$

$$P(Y=\text{t}) = 5/6$$

$$P(Y=\text{f}) = 1/6$$

$$\begin{aligned} H(Y) &= - 5/6 \log_2 5/6 - 1/6 \log_2 1/6 \\ &= 0.65 \end{aligned}$$



X_1	X_2	Y
T	T	T
T	F	T
T	T	T
T	F	T
F	T	T
F	F	F



Conditional Entropy

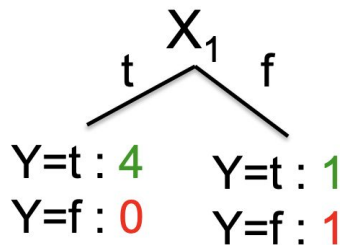
Conditional Entropy $H(Y|X)$ of a random variable Y conditioned on a random variable X

$$H(Y | X) = - \sum_{j=1}^v P(X = x_j) \sum_{i=1}^k P(Y = y_i | X = x_j) \log_2 P(Y = y_i | X = x_j)$$

Example:

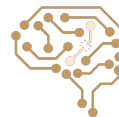
$$P(X_1 = \text{t}) = 4/6$$

$$P(X_1 = \text{f}) = 2/6$$



$$\begin{aligned} H(Y|X_1) &= - 4/6 (1 \log_2 1 + 0 \log_2 0) \\ &\quad - 2/6 (1/2 \log_2 1/2 + 1/2 \log_2 1/2) \\ &= 2/6 \end{aligned}$$

X_1	X_2	Y
T	T	T
T	F	T
T	T	T
T	F	T
F	T	T
F	F	F



Information Gain

- Decrease in entropy (uncertainty) after splitting

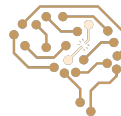
$$IG(X) = H(Y) - H(Y | X)$$

In our running example:

$$\begin{aligned} IG(X_1) &= H(Y) - H(Y|X_1) \\ &= 0.65 - 0.33 \end{aligned}$$

$IG(X_1) > 0 \rightarrow$ we prefer the split!

X_1	X_2	Y
T	T	T
T	F	T
T	T	T
T	F	T
F	T	T
F	F	F



Information Gain

- Information Gain (IG) or reduction in entropy from using attribute (feature) A:

$$IG(A) = Entropy\ before - AvgEntropy\ after\ choosing\ A$$

- Choose the attribute with the largest IG



Learning Decision Trees

- Start from empty decision tree
- Split on **next best attribute (feature)**
 - Use, for example, information gain to select attribute:

$$\arg \max_i IG(X_i) = \arg \max_i H(Y) - H(Y | X_i)$$

- Recurse



Suppose we want
to predict MPG

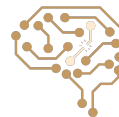
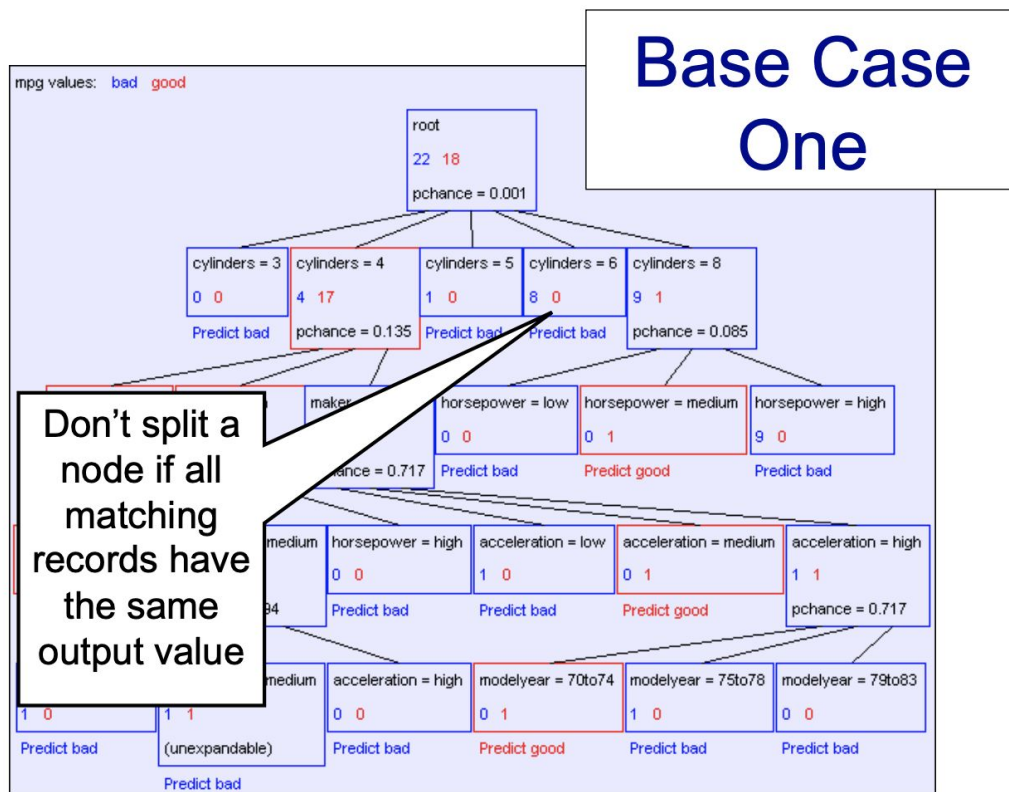
Look at all the
information
gains...

Information gains using the training set (40 records)
mpg values: bad good

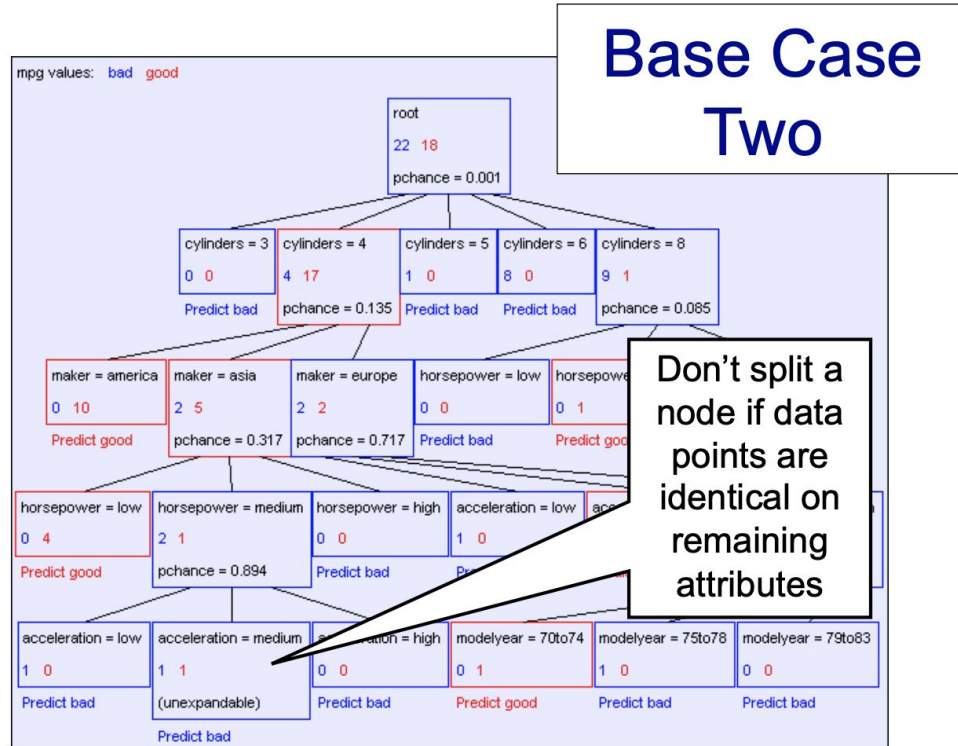
Input	Value	Distribution	Info Gain
cylinders	3		0.506731
	4		
	5		
	6		
	8		
displacement	low		0.223144
	medium		
	high		
horsepower	low		0.387605
	medium		
	high		
weight	low		0.304018
	medium		
	high		
acceleration	low		0.0642088
	medium		
	high		
modelyear	70to74		0.267964
	75to78		
	79to83		



When to terminate splitting?



When to terminate splitting?



When to terminate splitting?

- **Base Case One:** If all records in current data subset have the same output then **don't recurse**
- **Base Case Two:** If all records have exactly the same set of input attributes then **don't recurse**

Proposed Base Case 3:
If all attributes have zero
information gain then **don't
recurse**

•Is this a good idea?



When to terminate splitting?

$$y = a \text{ XOR } b$$

a	b	y
0	0	0
0	1	1
1	0	1
1	1	0

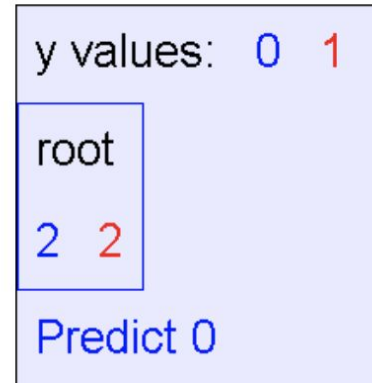
The information gains:

Information gains using the training set (4 records)

y values: 0 1

Input	Value	Distribution	Info Gain
a	0		0
	1		
b	0		0
	1		

The resulting decision tree:



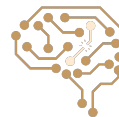
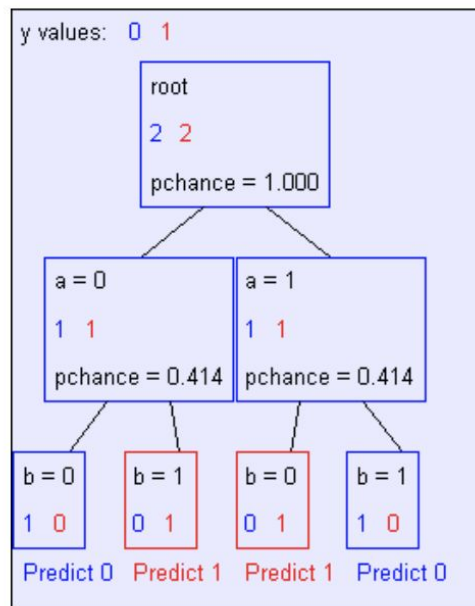
When to terminate splitting?

$y = a \text{ XOR } b$

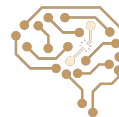
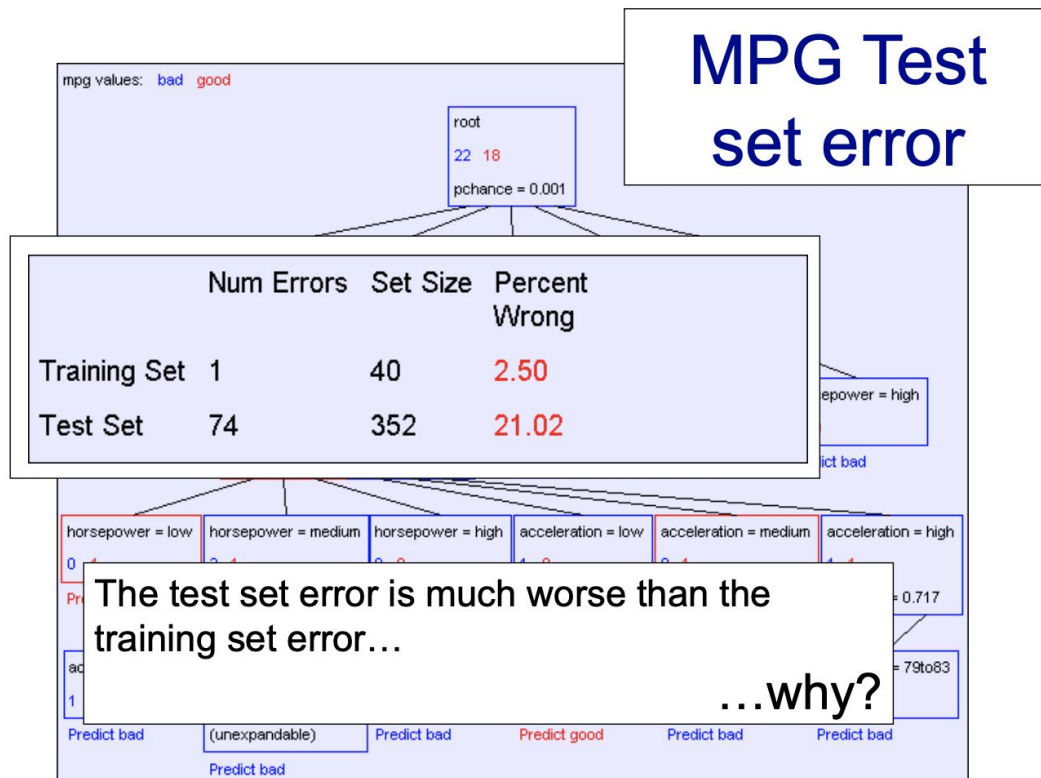
a	b	y
0	0	0
0	1	1
1	0	1
1	1	0

Is it OK to omit Base Case 3?

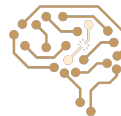
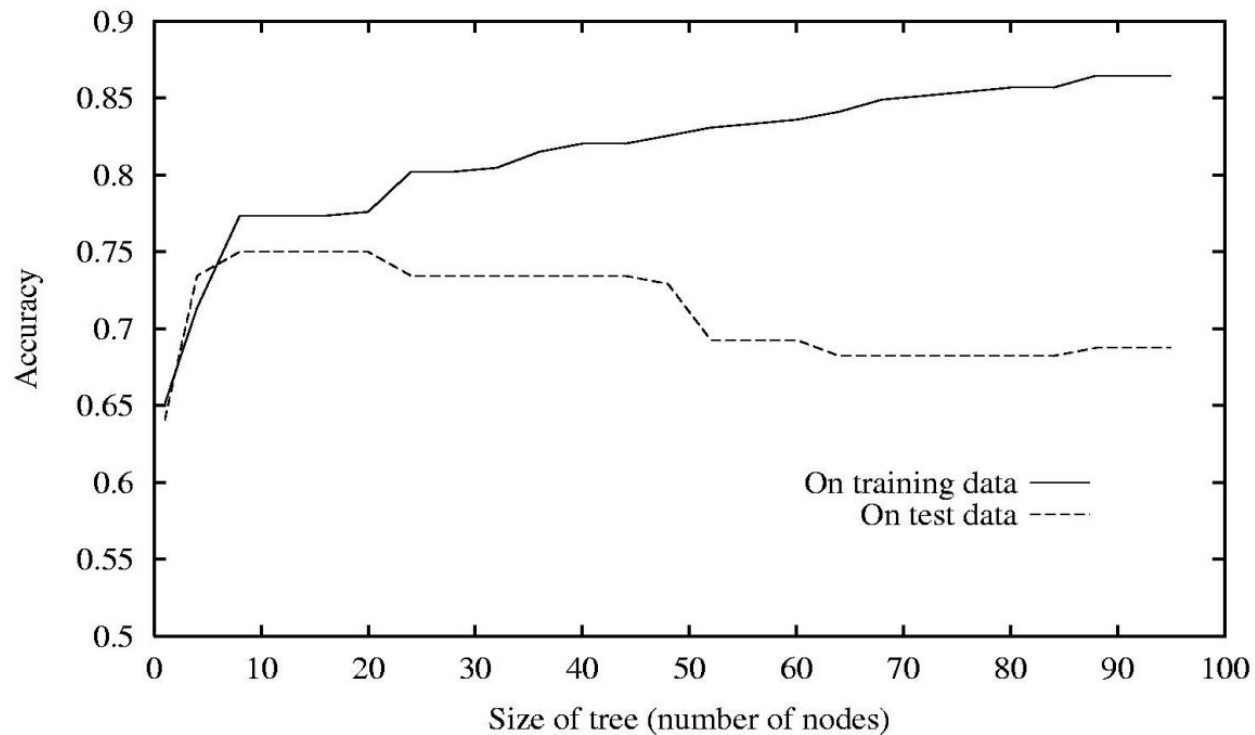
The resulting decision tree:



Decision Trees will Overfit!

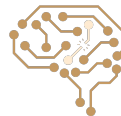


Decision Trees will Overfit!



Decision Trees will Overfit!

- Standard decision trees have no learning bias
 - Training set error is always zero!
 - (If there is no label noise)
 - Lots of variance
 - Must introduce some bias towards simpler trees
- Many strategies for picking simpler trees
 - Fixed depth
 - Fixed number of leaves
 - Or something smarter...



Next Session: Machine Learning Concepts

